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Inventor(s)	Cheng; Kuan-Hao et al.

Semiconductor device structure and method for forming the same

Abstract

A semiconductor device structure is provided. The semiconductor device structure includes a substrate having a base and a fin over the base. The semiconductor device structure includes a gate stack over a top portion of the fin. The semiconductor device structure includes a first nanostructure over the fin and passing through the gate stack. The semiconductor device structure includes a second nanostructure over the first nanostructure and passing through the gate stack. The first nanostructure is thicker than the second nanostructure. The semiconductor device structure includes a stressor structure over the fin and connected to the first nanostructure and the second nanostructure.

Inventors: **Cheng; Kuan-Hao (Hsinchu, TW), Lee; Wei-Yang (Taipei, TW), Chiu; Tzu-Hua (Hsinchu, TW), Fan; Wei-Han (Hsin-Chu, TW), Lin; Po-Yu (New Taipei, TW), Lin; Chia-Pin (Xinpu Township, Hsinchu County, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.** (Hsinchu, TW)

Family ID: **1000008782696**

Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD. (Hsinchu, TW)**

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Primary Examiner: Hanley; Britt D

Assistant Examiner: Boatman; Casey Paul

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

(2) In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

(3) However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1**A-**1**B are perspective views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

(3) FIGS. **2**A-**2**H are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

(4) FIG. **2**H-**1** is a perspective view of the semiconductor device structure of FIG. **2**H, in accordance with some embodiments.

(5) FIG. **2**H-**2** is a cross-sectional view illustrating the semiconductor device structure along a sectional line **2**H-**2**-**2**H-**2**′ in FIG. **2**H-**1**, in accordance with some embodiments.

(6) FIG. **3** is a cross-sectional view of a semiconductor device structure, in accordance with some embodiments.

(7) FIGS. **4**A-**4**D are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's

relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(10) The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. The term “substantially” may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term “substantially” may also relate to 90% of what is specified or higher, such as 95% of what is specified or higher, especially 99% of what is specified or higher, including 100% of what is specified, though the present invention is not limited thereto. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y.
(11) The term “about” may be varied in different technologies and be in the deviation range understood by the skilled in the art. The term “about” in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. For example, the term “about” may include deviations of up to 10% of what is specified, though the present invention is not limited thereto. The term “about” in relation to a numerical value x may mean x±5 or 10% of what is specified, though the present invention is not limited thereto.
(12) Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.
(13) The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.
(14) FIGS. **1**A-**1**B are perspective views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. As shown in FIG. **1**A, a substrate **110** is provided, in accordance with some embodiments. The substrate **110** includes, for example, a semiconductor substrate. The substrate **110** includes, for example, a semiconductor wafer (such as a silicon wafer) or a portion of a semiconductor wafer.
(15) In some embodiments, the substrate **110** is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure. In some other embodiments, the substrate **110** is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof. The substrate **110** may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such

as silicon on insulator or germanium on insulator), or a combination thereof.
(16) In some embodiments, the substrate **110** is a device wafer that includes various device elements. In some embodiments, the various device elements are formed in and/or over the substrate **110**. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate **110**. The passive devices include resistors, capacitors, or other suitable passive devices.
(17) For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc.
(18) Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.
(19) In some embodiments, isolation features (not shown) are formed in the substrate **110**. The isolation features are used to define active regions and electrically isolate various device elements formed in and/or over the substrate **110** in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.
(20) As shown in FIG. **1**A, a semiconductor layer **121′** is formed over the substrate **110**, in accordance with some embodiments. The semiconductor layer **121′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(21) As shown in FIG. **1**A, a semiconductor layer **122′** is formed over the semiconductor layer **121′**, in accordance with some embodiments. The semiconductor layer **122′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(22) As shown in FIG. **1**A, a semiconductor layer **123′** is formed over the semiconductor layer **122′**, in accordance with some embodiments. The semiconductor layer **123′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(23) As shown in FIG. **1**A, a semiconductor layer **124′** is formed over the semiconductor layer **123′**, in accordance with some embodiments. The semiconductor layer **124′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(24) As shown in FIG. **1**A, a semiconductor layer **125′** is formed over the semiconductor layer **124′**, in accordance with some embodiments. The semiconductor layer **125′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(25) As shown in FIG. **1**A, a semiconductor layer **126′** is formed over the semiconductor layer **125′**, in accordance with some embodiments. The semiconductor layer **126′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(26) As shown in FIG. **1**A, a semiconductor layer **127′** is formed over the semiconductor layer **126′**, in accordance with some embodiments. The semiconductor layer **127′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.
(27) As shown in FIG. **1**A, a semiconductor layer **128′** is formed over the semiconductor layer **127′**,

in accordance with some embodiments. The semiconductor layer **128′** is formed using an epitaxial process or a deposition process, such as a chemical vapor deposition process or a physical vapor deposition process, in accordance with some embodiments.

(28) The semiconductor layer **122′** is thicker than the semiconductor layer **121′**, **123′**, **124′**, **125′**, **126′**, **127′**, or **128′**, in accordance with some embodiments. The semiconductor layer **121′** or **123′** is thinner than the semiconductor layer **124′**, **125′**, **126′**, **127′**, or **128′**, in accordance with some embodiments.

(29) The semiconductor layers **121**, **123′**, **125′**, and **127′** are made of a same first material, in accordance with some embodiments. The first material is different from the material of the substrate **110**, in accordance with some embodiments. The first material includes an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure, in accordance with some embodiments.

(30) The first material includes a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof, in accordance with some embodiments.

(31) The semiconductor layers **122′**, **124′**, **126′**, and **128′** are made of a same second material, in accordance with some embodiments. The second material is different from the first material, in accordance with some embodiments. The second material is the same as the material of the substrate **110**, in accordance with some embodiments.

(32) The second material includes an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure, in accordance with some embodiments. The second material includes a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof, in accordance with some embodiments.

(33) As shown in FIG. **1**A, a mask layer **10** is formed over the semiconductor layer **128′**, in accordance with some embodiments. The mask layer **10** is made of a nitride-containing material, such as silicon nitride, a polymer material, such as a photoresist material, an oxide-containing material, such as silicon dioxide (SiO.sub.2), or another suitable material, which is different from the materials of the semiconductor layers **121′**, **122′**, **123′**, **124′**, **125′**, **126′**, **127′**, and **128′** and the substrate **110**, in accordance with some embodiments.

(34) The mask layer **10** is formed using a deposition process (e.g., a physical vapor deposition process or a chemical vapor deposition process), a photolithography process, and an etching process (e.g., a dry etching process), in accordance with some embodiments.

(35) FIG. **2**A is a cross-sectional view illustrating the semiconductor device structure along a sectional line **2**A-**2**A′ in FIG. **1**B, in accordance with some embodiments. As shown in FIGS. **1**A, **1**B and **2**A, the semiconductor layers **121′**, **122′**, **123′**, **124′**, **125′**, **126′**, **127′**, and **128′** and the substrate **110**, which are not covered by the mask layer **10**, are removed, in accordance with some embodiments.

(36) The removal process forms trenches **111** in the substrate **110**, in accordance with some embodiments. After the removal process, the remaining portion of the substrate **110** has a base **112** and a fin **114** over the base **112**, in accordance with some embodiments. The fin **114** is between the trenches **111**, in accordance with some embodiments.

(37) After the removal process, the remaining semiconductor layers **121′**, **122′**, **123′**, **124′**, **125′**, **126′**, **127′**, and **128′** respectively form nanostructures **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128**, in accordance with some embodiments. The nanostructures **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128** together form a nanostructure stack **120**, in accordance with some embodiments.

(38) The nanostructure stack **120** is formed over the fin **114**, in accordance with some embodiments. In some embodiments, as shown in FIG. **2**A, a thickness T**120** of the nanostructure stack **120** ranges from about 40 nm to about 80 nm.

(39) The nanostructures **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128** are sequentially stacked over the fin **114**, in accordance with some embodiments. The nanostructures **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128** include nanowires or nanosheets, in accordance with some embodiments.
(40) The nanostructure **122** is thicker than the nanostructure **121**, **123**, **124**, **125**, **126**, **127**, or **128**, in accordance with some embodiments. The nanostructure **121** or **123** is thinner than the nanostructure **125** or **127**, in accordance with some embodiments. The nanostructure **122** is closer to the nanostructure **124** than the nanostructure **126**, in accordance with some embodiments.
(41) The nanostructure **121** has a thickness T**121** ranging from about 1 nm to about 5 nm, in accordance with some embodiments. The nanostructure **122** has a thickness T**122** ranging from about 7 nm to about 20 nm, in accordance with some embodiments. The nanostructure **123** has a thickness T**123** ranging from about 1 nm to about 5 nm, in accordance with some embodiments. The nanostructure **124** has a thickness T**124** ranging from about 4 nm to about 14 nm, in accordance with some embodiments. The nanostructure **125** has a thickness T**125** ranging from about 4 nm to about 10 nm, in accordance with some embodiments.
(42) The nanostructure **126** has a thickness T**126** ranging from about 4 nm to about 14 nm, in accordance with some embodiments. The nanostructure **127** has a thickness T**127** ranging from about 4 nm to about 10 nm, in accordance with some embodiments. The nanostructure **128** has a thickness T**128** ranging from about 4 nm to about 14 nm, in accordance with some embodiments. The removal process includes an anisotropic etching process, such as a dry etching process (e.g., a plasma etching process), in accordance with some embodiments.
(43) As shown in FIG. **1**B, the mask layer **10** is removed, in accordance with some embodiments. As shown in FIG. **1**B, an isolation layer **130** is formed over the base **112**, in accordance with some embodiments. The fin **114** is partially embedded in the isolation layer **130**, in accordance with some embodiments. The fin **114** is surrounded by the isolation layer **130**, in accordance with some embodiments.
(44) The isolation layer **130** is made of a dielectric material such as an oxide-containing material (e.g., silicon oxide), an oxynitride-containing material (e.g., silicon oxynitride), a low-k (low dielectric constant) material, a porous dielectric material, glass, or a combination thereof, in accordance with some embodiments. The glass includes borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), or a combination thereof, in accordance with some embodiments.
(45) The isolation layer **130** is formed using a deposition process (or a spin-on process), a chemical mechanical polishing process, and an etching back process, in accordance with some embodiments. The deposition process includes a chemical vapor deposition (CVD) process, a high density plasma chemical vapor deposition (HDPCVD) process, a flowable chemical vapor deposition (FCVD) process, a sputtering process, or a combination thereof, in accordance with some embodiments.
(46) As shown in FIGS. **1**B and **2**A, gate stacks **140** are formed over the nanostructure stack **120**, the fin **114**, and the isolation layer **130**, in accordance with some embodiments. For the sake of simplicity, FIG. **1**B only shows one of the gate stacks **140**, in accordance with some embodiments. As shown in FIG. **2**A, trenches T are between the adjacent gate stacks **140** to separate the adjacent gate stacks **140** from one another, in accordance with some embodiments.
(47) The nanostructures **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128** pass through the gate stack **140**, in accordance with some embodiments. Each gate stack **140** includes a gate dielectric layer **142** and a gate electrode **144**, in accordance with some embodiments. The gate electrode **144** is over the gate dielectric layer **142**, in accordance with some embodiments. The gate dielectric layer **142** is positioned between the gate electrode **144** and the nanostructure stack **120**, in accordance with some embodiments.
(48) The gate dielectric layer **142** is also positioned between the gate electrode **144** and the fin **114**, in accordance with some embodiments. The gate dielectric layer **142** is positioned between the gate electrode **144** and the isolation layer **130**, in accordance with some embodiments.

(49) The gate dielectric layer **142** is made of an oxide-containing material such as silicon oxide, in accordance with some embodiments. The gate dielectric layer **142** is formed using a chemical vapor deposition process and an etching process, in accordance with some embodiments. The gate electrode **144** is made of a semiconductor material such as polysilicon, in accordance with some embodiments. The gate electrode **144** is formed using a chemical vapor deposition process and an etching process, in accordance with some embodiments.
(50) As shown in FIGS. **1**B and **2**A, a mask layer **150** is formed over the gate stacks **140**, in accordance with some embodiments. The mask layer **150** is made of a material different from the materials of the gate stacks **140**, in accordance with some embodiments. The mask layer **150** is made of nitrides (e.g., silicon nitride) or oxynitride (e.g., silicon oxynitride), in accordance with some embodiments.
(51) As shown in FIGS. **1**B and **2**A, a spacer structure **160** is formed over sidewalls **142***a*. **144***a* and **152** of the gate dielectric layer **142**, the gate electrode **144** and the mask layer **150**, in accordance with some embodiments. The spacer structure **160** surrounds the gate stack **140** and the mask layer **150**, in accordance with some embodiments. The spacer structure **160** is positioned over the nanostructure stack **120**, the fin **114** and the isolation layer **130**, in accordance with some embodiments.
(52) As shown in FIG. **2**A, the spacer structure **160** includes spacer layers **162** and **164**, in accordance with some embodiments. The spacer layer **162** is between the spacer layer **164** and the gate stack **140**, in accordance with some embodiments. The spacer layer **162** is also between the spacer layer **164** and the mask layer **150**, in accordance with some embodiments. The spacer layers **162** and **164** are made of different materials, in accordance with some embodiments.
(53) The spacer layers **162** and **164** include insulating materials, such as silicon oxide, silicon nitride, silicon oxynitride, or silicon carbide, in accordance with some embodiments. The spacer layers **162** and **164** are made of materials different from that of the gate stack **140** and the mask layer **150**, in accordance with some embodiments. The formation of the spacer layers **162** and **164** includes deposition processes and an anisotropic etching process, in accordance with some embodiments.
(54) FIGS. **2**A-**2**H are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. After the step of FIG. **2**A, as shown in FIG. **2**B, portions of the nanostructure stack **120**, which are not covered by the gate stacks **140** and the spacer structure **160**, are removed, in accordance with some embodiments. The removal process forms trenches **120***a* in the nanostructure stack **120** and the fin **114**, in accordance with some embodiments.
(55) Each trench **120***a* has a width W**120***a* decreasing toward the fin **114**, in accordance with some embodiments. The width W**120***a* is measured along a longitudinal axis **114***a* of the fin **114**, in accordance with some embodiments. The widths shown in FIGS. **2**B and **2**E are also measured along the longitudinal axis **114***a* of the fin **114**, in accordance with some embodiments.
(56) The removal process includes an etching process, in accordance with some embodiments. The etching process includes an anisotropic etching process such as a dry etching process, in accordance with some embodiments.
(57) As shown in FIG. **2**C, end portions of the nanostructures **121**, **123**, **125** and **127** are removed through the trenches **120***a* and T, in accordance with some embodiments. The removal process forms recesses R**1**, R**2**, R**3** and R**4** in the nanostructure stack **120**, in accordance with some embodiments. The recess R**1** is between the fin **114** and the nanostructure **122**, in accordance with some embodiments.
(58) The recess R**2** is between the nanostructures **122** and **124**, in accordance with some embodiments. The recess R**3** is between the nanostructures **124** and **126**, in accordance with some embodiments. The recess R**4** is between the nanostructures **126** and **128**, in accordance with some embodiments. The removal process includes an etching process such as a dry etching process or a

wet etching process, in accordance with some embodiments.

(59) As shown in FIG. **2**C, an inner spacer material layer **170** is formed over the mask layer **150**, the spacer structure **160**, the nanostructure stack **120** and the fin **114**, in accordance with some embodiments. The recesses R**1**, R**2**, R**3** and R**4** are filled with the inner spacer material layer **170**, in accordance with some embodiments. The inner spacer material layer **170** is in direct contact with sidewalls **121***a*, **123***a*, **125***a* and **127***a* of the nanostructures **121**, **123**, **125** and **127**, in accordance with some embodiments.

(60) The inner spacer material layer **170** is made of an insulating material, such as an oxide-containing material (e.g., silicon oxide), a nitride-containing material (e.g., silicon nitride), an oxynitride-containing material (e.g., silicon oxynitride), a carbide-containing material (e.g., silicon carbide), a high-k material (e.g., HfO.sub.2, ZrO.sub.2, HfZrO.sub.2, or Al.sub.2O.sub.3), or a low-k material, in accordance with some embodiments.

(61) The term "high-k material" means a material having a dielectric constant greater than the dielectric constant of silicon dioxide, in accordance with some embodiments. The term "low-k material" means a material having a dielectric constant less than the dielectric constant of silicon dioxide, in accordance with some embodiments. The inner spacer material layer **170** is formed using a deposition process such as a physical vapor deposition process, a chemical vapor deposition process, an atomic layer deposition process, or the like, in accordance with some embodiments.

(62) As shown in FIG. **2**D, the portions of the inner spacer material layer **170** outside of the recesses R**1**, R**2**, R**3** and R**4** are partially removed, in accordance with some embodiments. The remaining inner spacer material layer **170** includes inner spacers **172**, **174**, **176** and **178** and an insulating layer **179**, in accordance with some embodiments. The inner spacers **172**, **174**, **176** and **178** are in the recesses R**1**, R**2**, R**3** and R**4** respectively, in accordance with some embodiments. The insulating layer **179** is over a bottom surface **120***al* of each trench **120***a*, in accordance with some embodiments.

(63) In some embodiments, sidewalls **172***a*, **174***a*, **176***a*, and **178***a* of the inner spacers **172**, **174**, **176**, and **178** are concave curved surfaces, in accordance with some embodiments. The bottom surface **120***al* is a concave curved surface, in accordance with some embodiments. In some embodiments, a top surface **179***a* of the insulating layer **179** is a concave curved surface, in accordance with some embodiments.

(64) As shown in FIG. **2**E, a lightly doped layer **182** is formed in the trenches **120***a* and over the insulating layer **179**, in accordance with some embodiments. The lightly doped layer **182** covers sidewalls **122***a*, **124***a*, **126***a*, **128***a*, **172***a*, **174***a*, **176***a*, and **178***a* of the nanostructures **122**, **124**, **126**, and **128** and the inner spacers **172**, **174**, **176**, and **178**, in accordance with some embodiments.

(65) The lightly doped layer **182** is in direct contact with the nanostructures **122**, **124**, **126** and **128**, the inner spacers **172**, **174**, **176** and **178**, and the substrate **110**, in accordance with some embodiments. The lightly doped layer **182** is formed using an epitaxial process, in accordance with some embodiments.

(66) Since the lightly doped layer **182** epitaxially grows from semiconductor surfaces of the fin **114** and the nanostructures **122**, **124**, **126**, and **128**, the lightly doped layer **182** has thick portions **182***a*, **182***b*, **182***c*, and **182***d*, which respectively grow from the nanostructures **122**, **124**, **126**, and **128**, in accordance with some embodiments.

(67) Since the lightly doped layer **182** does not grow from the inner spacers **174**, **176**, and **178**, the lightly doped layer **182** has thin merge portions **182***c*, **182***f*, and **182***g*, which are respectively connected to the inner spacers **174**, **176**, and **178**, in accordance with some embodiments. The thick portion **182***a*, **182***b*, **182***c*, or **182***d* is thicker than the thin merge portion **182***e*, **182***f*, or **182***g*, in accordance with some embodiments.

(68) As shown in FIG. **2**E, a heavily doped structure **184** is formed over the lightly doped layer **182**, in accordance with some embodiments. The lightly doped layer **182** and the heavily doped structure **184** are doped with the same dopant, in accordance with some embodiments.

(69) The lightly doped layer **182** separates the heavily doped structure **184** from the nanostructures **122**, **124**, **126**, and **128** and the fin **114**, in accordance with some embodiments. The lightly doped layer **182** surrounds a lower portion of the heavily doped structure **184**, in accordance with some embodiments. The lightly doped layer **182** and the heavily doped structure **184** in the same trench **120***a* together form a stressor structure **180**, in accordance with some embodiments. The stressor structures **180** are also referred to as source/drain structures, in accordance with some embodiments.
(70) In some embodiments, a concentration of the dopant in the lightly doped layer **182** is less than a concentration of the dopant in the heavily doped structure **184**. The concentration of the dopant in the lightly doped layer **182** is greater than the concentration of the dopant in the fin **114** and less than the concentration of the dopant in the heavily doped structure **184**, in accordance with some embodiments. Therefore, the lattice constant of the lightly doped layer **182** is between the lattice constant of the fin **114** and the lattice constant of the heavily doped structure **184**, in accordance with some embodiments. The formation of the lightly doped layer **182** helps the heavily doped structure **184** epitaxially grow over the fin **114**, in accordance with some embodiments.
(71) If the stressor structure **180** is used as a stressor structure of a p-channel field effect transistor (pFET), the dopant includes an element, which is larger than silicon, in accordance with some embodiments. The element includes germanium (Ge), in accordance with some embodiments. Since the element is larger than silicon, the stressor structures **180** are able to provide a compressive stress to the nanostructures **122**, **124**, **126** and **128** so as to improve the channel property of the nanostructures **122**, **124**, **126** and **128**, in accordance with some embodiments.
(72) The germanium concentration of the lightly doped layer **182** is less than the germanium concentration of the heavily doped structure **184**, in accordance with some embodiments. In some embodiments, a germanium concentration ratio of the heavily doped structure **184** to the lightly doped layer **182** ranges from about 1.5 to about 2.5.
(73) The germanium concentration of the lightly doped layer **182** ranges from about 15 at. % to about 30 at. % (atomic percent), in accordance with some embodiments. The germanium concentration of the heavily doped structure **184** ranges from about 35 at. % to about 55 at. % (atomic percent), in accordance with some embodiments.
(74) If the stressor structure **180** is used as a stressor structure of an n-channel field effect transistor (nFET), the dopant includes an element, which is smaller than silicon, in accordance with some embodiments. The element includes carbon (C), in accordance with some embodiments. Since the element is smaller than silicon, the stressor structures **180** are able to provide a tensile stress to the nanostructures **122**, **124**, **126** and **128** so as to improve the channel property of the nanostructures **122**, **124**, **126** and **128**, in accordance with some embodiments.
(75) The carbon concentration of the lightly doped layer **182** is less than the carbon concentration of the heavily doped structure **184**, in accordance with some embodiments. In some embodiments, a carbon concentration ratio of the heavily doped structure **184** to the lightly doped layer **182** ranges from about 1.5 to about 2.5.
(76) The carbon concentration of the lightly doped layer **182** ranges from about 0.1 at. % to about 10 at. % (atomic percent), in accordance with some embodiments. In some other embodiments, the lightly doped layer **182** has no carbon. The carbon concentration of the heavily doped structure **184** ranges from about 10 at. % to about 20 at. % (atomic percent), in accordance with some embodiments.
(77) Since the concentration of the dopant in the heavily doped structure **184** is greater than the concentration of the dopant in the lightly doped layer **182**, the heavily doped structure **184** is able to provide more stress to the nanostructures **122**, **124**, **126** and **128** than the lightly doped layer **182**, in accordance with some embodiments. Therefore, if the lateral overlap area between the nanostructures **122**, **124**, **126** and **128** and the heavily doped structures **184** is increased, the heavily doped structures **184** is able to provide more stress to the nanostructures **122**, **124**, **126** and **128**.

(78) Since the application increases the thickness T**122** of the nanostructure **122**, the height of the nanostructure **122** is increased, which increases the lateral overlap area between the nanostructure **122** and the heavily doped structures **184**, in accordance with some embodiments. Therefore, the stress applied to the nanostructure **122** from the heavily doped structures **184** is increased, in accordance with some embodiments. As a result, the channel property of the nanostructure **122** is improved (or boosted), in accordance with some embodiments.
(79) In some embodiments, a region R of the stressor structure **180** is lower than a top surface **122***b* of the nanostructure **122** and higher than a bottom surface **122***c* of the nanostructure **122**. In the region R, the heavily doped structure **184** is thicker than a portion **182***h* of the lightly doped layer **182** under the heavily doped structure **184**, in accordance with some embodiments. That is, in the region R, the thickness T**184***a* of the heavily doped structure **184** is greater than the thickness T**182***h* of the portion **182***h* of the lightly doped layer **182**, in accordance with some embodiments.
(80) In the region R of the stressor structure **180**, the thick portion **182***a* of the lightly doped layer **182** between the heavily doped structure **184** and the nanostructure **122** is thicker than the portion **182***h* of the lightly doped layer **182** under the heavily doped structure **184**, in accordance with some embodiments.
(81) The thick portions **182***a* and **182***b* of the lightly doped layer **182** respectively cover the sidewalls **122***a* and **124***a* of the nanostructures **122** and **124**, in accordance with some embodiments. The thick portion **182***a* is thicker than the thick portion **182***b*, in accordance with some embodiments.
(82) In some embodiments, a ratio of the thickness T**122** of the nanostructure **122** to the thickness T**124**, T**126**, or T**128** of the nanostructure **124**, **126**, or **128** ranges from about 1.5 to about 2. If the ratio (T**122**/T**124**) is less than 1.5, the lateral overlap area between the nanostructure **122** and the heavily doped structures **184** is not large enough to improve the channel property of the nanostructure **122**, in accordance with some embodiments. If the ratio (T**122**/T**124**) is greater than 2, the nanostructure **122** is too thick to latch the channel of the nanostructure **122**, in accordance with some embodiments.
(83) Since the application increases the thickness T**122** of the nanostructure **122**, a distance D**1** between the nanostructure **122** and the fin **114** and the distance D**2** between the nanostructures **122** and **124** are both decreased, in accordance with some embodiments. In some embodiments, the distance D**1** or D**2** is less than a distance D**3** between the nanostructures **124** and **126**. The distance D**1** or D**2** is less than the distance D**4** between the nanostructures **126** and **128**, in accordance with some embodiments.
(84) Since the distance D**1** is decreased, the thick portion **182***a* of the lightly doped layer **182**, which epitaxially grows from the sidewall **122***a* of the nanostructure **122**, easily merges with the lightly doped layer **182**, which epitaxially grows from the fin **114**, in accordance with some embodiments.
(85) Since the distance D**2** is decreased, the thick portion **182***a* of the lightly doped layer **182** easily merges with the thick portion **182***b* of the lightly doped layer **182**, which epitaxially grows from the sidewall **124***a* of the nanostructure **124**, in accordance with some embodiments.
(86) As a result, the increase in the thickness T**122** of the nanostructure **122** decreases the distances D**1** and D**2**, which helps the epitaxial growth of the lightly doped layer **182**, in accordance with some embodiments.
(87) The distance D**1** is substantially equal to the distance D**2**, in accordance with some embodiments. The thickness T**122**, T**124**, T**126**, or T**128** of the nanostructure **122**, **124**, **126**, or **128** is greater than the distance D**1**, D**2**, D**3**, or D**4**, in accordance with some embodiments.
(88) If the stressor structures **180** are used as stressor structures of a p-channel field effect transistor (pFET), the stressor structures **180** are made of a semiconductor material (e.g., silicon germanium) with P-type dopants, such as the Group IIIA element, in accordance with some embodiments. The Group IIIA element includes boron or another suitable material.

(89) If the stressor structures **180** are used as stressor structures of a n-channel field effect transistor (nFET), the stressor structures **180** are made of a semiconductor material (e.g., silicon or silicon carbide) with N-type dopants, such as the Group VA element, in accordance with some embodiments. The Group VA element includes phosphor (P), antimony (Sb), or another suitable Group VA material.
(90) As shown in FIG. **2**F, a cap layer **190** is formed over the stressor structures **180**, in accordance with some embodiments. The cap layer **190** is doped with the dopant, which is the same as the dopant in the lightly doped layer **182** and the heavily doped structure **184**, in accordance with some embodiments.
(91) The dopant concentration of the cap layer **190** is greater than the dopant concentration of the heavily doped structure **184**, in accordance with some embodiments. The cap layer **190** is used to prevent the dopant in the heavily doped structure **184** from diffusing into a dielectric layer, which is formed over the cap layer **190** in a subsequent process, in accordance with some embodiments.
(92) In some embodiments, the dopant includes germanium (Ge) or carbon (C). The dopant concentration of the cap layer **190** ranges from about 55 at. % to about 70 at. % (atomic percent), in accordance with some embodiments. The cap layer **190** includes a semiconductor material, such as silicon, in accordance with some embodiments. The cap layer **190** is made of silicon germanium or silicon carbide, in accordance with some embodiments.
(93) As shown in FIG. **2**F, a protection layer **210** is formed over the cap layer **190**, in accordance with some embodiments. The protection layer **210** is used to prevent the dopant in the cap layer **190** and the heavily doped structure **184** from diffusing into the dielectric layer, which is formed over the protection layer **210** in the subsequent process, in accordance with some embodiments. The protection layer **210** is made of a semiconductor material, such as pure silicon, in accordance with some embodiments.
(94) As shown in FIG. **2**F, a dielectric layer **220** is formed over the protection layer **210**, in accordance with some embodiments. The dielectric layer **220** includes a dielectric material such as an oxide-containing material (e.g., silicon oxide), an oxynitride-containing material (e.g., silicon oxynitride), a low-k material, a porous dielectric material, glass, or a combination thereof, in accordance with some embodiments.
(95) The glass includes borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), or a combination thereof, in accordance with some embodiments. The dielectric layer **220** is formed by a deposition process (e.g., a chemical vapor deposition process) and a planarization process (e.g., a chemical mechanical polishing process), in accordance with some embodiments.
(96) As shown in FIGS. **2**F and. **2**G, the gate stacks **140** and the mask layer **150** are removed, in accordance with some embodiments. The removal process forms a trench **166** in the spacer structure **160**, in accordance with some embodiments. As shown in FIGS. **2**F and **2**G, the nanostructures **121**, **123**, **125** and **127** are removed through the trench **166**, in accordance with some embodiments. The removal process for removing the gate stacks **140**, the mask layer **150** and the nanostructures **121**, **123**, **125** and **127** includes an etching process such as a wet etching process or a dry etching process, in accordance with some embodiments.
(97) As shown in FIG. **2**G, a gate dielectric layer **232** is formed over the dielectric layer **220**, the spacer structure **160**, the fin **114**, and the nanostructures **122**, **124**, **126** and **128**, in accordance with some embodiments. The gate dielectric layer **232** conformally covers the nanostructures **122**, **124**, **126** and **128** and inner walls and a bottom surface of the trench **166**, in accordance with some embodiments.
(98) The gate dielectric layer **232** is made of a high-K material, such as HfO.sub.2, ZrO.sub.2, HfZrO.sub.2, or Al.sub.2O.sub.3. The gate dielectric layer **232** is formed using an atomic layer deposition process or another suitable process.
(99) As shown in FIG. **2**G, a work function metal layer **234** is formed over the gate dielectric layer

232, in accordance with some embodiments. The work function metal layer **234** is conformally formed over the gate dielectric layer **232**, in accordance with some embodiments. The work function metal layer **234** is made of TiN, TaN, TiSiN, or another suitable conductive material. The work function metal layer **234** is formed using an atomic layer deposition process or another suitable process.
(100) As shown in FIG. **2**G, a gate electrode layer **236** is formed over the work function metal layer **234**, in accordance with some embodiments. The gate electrode layer **236** is made of W, Co, Al, or another suitable conductive material. The gate electrode layer **236** is formed using an atomic layer deposition process or another suitable process.
(101) FIG. **2**H-**1** is a perspective view of the semiconductor device structure of FIG. **2**H, in accordance with some embodiments. FIG. **2**H is a cross-sectional view illustrating the semiconductor device structure along a sectional line **2**H-**2**H′ in FIG. **2**H-**1**, in accordance with some embodiments. FIG. **2**H-**2** is a cross-sectional view illustrating the semiconductor device structure along a sectional line **2**H-**2**-**2**H-**2**′ in FIG. **2**H-**1**, in accordance with some embodiments.
(102) As shown in FIGS. **2**H, **2**H-**1**, and **2**H-**2**, the gate dielectric layer **232**, the work function metal layer **234**, and the gate electrode layer **236** outside of the trench **166** are removed, in accordance with some embodiments. The removal process includes a chemical mechanical polishing process, in accordance with some embodiments.
(103) After the removal process, the gate dielectric layer **232**, the work function metal layer **234**, the gate electrode layer **236** remaining in the same trench **166** together form a gate stack **230**, in accordance with some embodiments. In this step, a semiconductor device structure **100** is substantially formed, in accordance with some embodiments. As shown in FIG. **2**H-**2**, the gate stack **230** surrounds the nanostructures **122**, **124**, **126** and **128**, in accordance with some embodiments.
(104) As shown in FIG. **2**H, the gate stack **230** has portions **231**, **233**, **235**, and **237**, in accordance with some embodiments. The portion **231** is between and in direct contact with the nanostructure **122** and the fin **114**, in accordance with some embodiments. The portion **233** is between and in direct contact with the nanostructures **122** and **124**, in accordance with some embodiments.
(105) The portion **235** is between and in direct contact with the nanostructures **124** and **126**, in accordance with some embodiments. The portion **237** is between and in direct contact with the nanostructures **126** and **128**, in accordance with some embodiments.
(106) The portions **231**, **233**, **235**, and **237** respectively have thicknesses T**231**, T**233**, T**235**, and T**237**, in accordance with some embodiments. The portion **231** or **233** is thinner than the portion **235** or **237**, in accordance with some embodiments. That is, the thickness T**231** or T**233** is less than the thickness T**235** or T**237**, in accordance with some embodiments.
(107) The portions **231** and **233** have a substantially same thickness, in accordance with some embodiments. That is, the thicknesses T**231** is substantially equal to the thicknesses T**233**, in accordance with some embodiments. The portions **235** and **237** have a substantially same thickness, in accordance with some embodiments. That is, the thicknesses T**235** is substantially equal to the thicknesses T**237**, in accordance with some embodiments. The portion **235** is closer to the portion **233** than the portion **231**, in accordance with some embodiments.
(108) The nanostructures **122**, **124**, **126** and **128** pass through the gate stack **230**, in accordance with some embodiments. The nanostructure **122** is wider than the nanostructure **124**, in accordance with some embodiments.
(109) The stressor structure **180** adjacent to the nanostructure **122** is narrower than the stressor structure **180** adjacent to the nanostructure **124**, in accordance with some embodiments. The fin **114** and the nanostructures **122**, **124**, **126** and **128** are spaced apart from each other, in accordance with some embodiments.
(110) The portion **182***al* of the lightly doped layer **182** is between the heavily doped structure **184** and the nanostructure **122**, in accordance with some embodiments. The portion **182***al* covers an

upper portion A of the sidewall **122***a* of the nanostructure **122**, in accordance with some embodiments.

(111) In some embodiments, a lower portion B of the sidewall **122***a* is covered by a portion **182***a***2** of the lightly doped layer **182**, which is not between the heavily doped structure **184** and the nanostructure **122**. The upper portion A is greater than the lower portion B, in accordance with some embodiments. For example, the length LA of the upper portion A is greater than the length L.sub.B of the lower portion B.

(112) Since the thickness T**122** of the nanostructure **122** is increased, the upper portion A is enlarged, in accordance with some embodiments. The stress from the heavily doped structure **184** is laterally applied to the upper portion A of the sidewall **122***a* of the nanostructure **122**, in accordance with some embodiments. Therefore, the increase in the thickness T**122** of the nanostructure **122** increases the stress applied to nanostructure **122** from the heavily doped structure **184**, in accordance with some embodiments. As a result, the channel property of the nanostructure **122** is improved or boosted, in accordance with some embodiments. Therefore, the performance of the semiconductor device structure **100** is improved or boosted, in accordance with some embodiments.

(113) FIG. **3** is a cross-sectional view of a semiconductor device structure **300**, in accordance with some embodiments. As shown in FIG. **3**, the semiconductor device structure **300** is similar to the semiconductor device structure **100** of FIG. **2**H, except that the distance D**1** between the nanostructure **122** and the fin **114**, the distance D**2** between the nanostructures **122** and **124**, the distance D**3** between the nanostructures **124** and **126**, and the distance D**4** between the nanostructures **126** and **128** are substantially equal to each other, in accordance with some embodiments.

(114) FIGS. **4**A-**4**D are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. After the step of FIG. **2**C, as shown in FIG. **4**A, the portions of the inner spacer material layer **170** outside of the recesses R**1**, R**2**, R**3** and R**4** are partially removed, in accordance with some embodiments. The remaining inner spacer material layer **170** includes inner spacers **172**, **174**, **176** and **178** but no insulating layer **179**, in accordance with some embodiments.

(115) After the removal process, the bottom surface **120***al* of each trench **120***a* is exposed, in accordance with some embodiments. The bottom surface **120***al* is a (100) surface, in accordance with some embodiments. The (100) surface is a (100) surface crystal orientation of the material (e.g., Si) of the substrate **110**, in accordance with some embodiments.

(116) In some embodiments, inner walls **120***a***2** of each trench **120***a* are (110) surfaces, in accordance with some embodiments. The inner walls **120***a***2** include the sidewalls **122***a*, **124***a*, **126***a*, and **128***a* of the nanostructures **122**, **124**, **126**, and **128**, in accordance with some embodiments. The (110) surface is a (110) surface crystal orientation of the material (e.g., Si) of the substrate **110**, in accordance with some embodiments.

(117) As shown in FIG. **4**B, a semiconductor layer **410** is formed over the bottom surface **120***al* and the inner walls **120***a***2** of each trench **120***a*, in accordance with some embodiments. The semiconductor layer **410** is made of a semiconductor material, such as silicon, in accordance with some embodiments. The semiconductor layer **410** is formed using an epitaxial process, in accordance with some embodiments.

(118) Since the epitaxial growth rate of the semiconductor layer **410** over the (100) surface is greater than the epitaxial growth rate of the semiconductor layer **410** over the (110) surface, the semiconductor layer **410** has a thick portion **412** and a thin portion **414**, in accordance with some embodiments. The thick portion **412** is over the bottom surface **120***al* of each trench **120***a*, in accordance with some embodiments. The thin portion **414** is over the inner walls **120***a***2** of each trench **120***a*, in accordance with some embodiments.

(119) As shown in FIG. **4**C, the thin portion **414** of the semiconductor layer **410** is removed, in

accordance with some embodiments. The removal process includes an etching process, such as a wet etching process or a dry etching process, in accordance with some embodiments. The dry etching process is performed using an etching gas, such as an HCl-containing gas, in accordance with some embodiments.
(120) As shown in FIG. **4**D, the steps of FIGS. **4**B-**4**C are performed many times to increase the thickness of the thick portion **412**, in accordance with some embodiments. The thick portion **412** has a top surface **412***a*, in accordance with some embodiments. The top surface **412***a* is a flat top surface, in accordance with some embodiments. The top surface **412***a* is a (100) surface, in accordance with some embodiments.
(121) As shown in FIG. **4**D, the steps of FIGS. **2**E-**2**H are performed to form the stressor structures **180**, the cap layer **190**, the protection layer **210**, and the dielectric layer **220**, to remove the mask layer **150**, the gate stacks **140**, and the nanostructures **121**, **123**, **125** and **127**, and to form the gate stack **230**, in accordance with some embodiments. In this step, a semiconductor device structure **400** is substantially formed, in accordance with some embodiments.
(122) Processes and materials for forming the semiconductor device structures **300** and **400** may be similar to, or the same as, those for forming the semiconductor device structure **100** described above. Elements designated by the same reference numbers as those in FIGS. **1**A to **4**D have the structures and the materials similar thereto or the same thereas. Therefore, the detailed descriptions thereof will not be repeated herein.
(123) In accordance with some embodiments, semiconductor device structures and methods for forming the same are provided. The methods (for forming the semiconductor device structure) form a gate all around structure with a thicker nanostructure and a thinner nanostructure over the thicker nanostructure. The thicker nanostructure laterally overlaps with a larger part of a heavily doped structure of a stressor structure, which increases the stress applied to the thicker nanostructure from the heavily doped structure. Therefore, the channel property of the thicker nanostructure is improved, which improves the performance of the gate all around structure.
(124) In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a substrate having a base and a fin over the base. The semiconductor device structure includes a gate stack over a top portion of the fin. The semiconductor device structure includes a first nanostructure over the fin and passing through the gate stack. The semiconductor device structure includes a second nanostructure over the first nanostructure and passing through the gate stack. The first nanostructure is thicker than the second nanostructure. The semiconductor device structure includes a stressor structure over the fin and connected to the first nanostructure and the second nanostructure.
(125) In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a substrate having a base and a fin over the base. The semiconductor device structure includes a gate stack over a top portion of the fin. The semiconductor device structure includes a first nanostructure over the fin and passing through the gate stack. The semiconductor device structure includes a second nanostructure over the first nanostructure and passing through the gate stack. The semiconductor device structure includes a third nanostructure over the second nanostructure and passing through the gate stack. The gate stack has a first portion, a second portion, and a third portion, the first portion is between and in direct contact with the fin and the first nanostructure, the second portion is between and in direct contact with the first nanostructure and the second nanostructure, the third portion is between and in direct contact with the second nanostructure and the third nanostructure, the third portion is closer to the second portion than the first portion. The semiconductor device structure includes a stressor structure over the fin and connected to the first nanostructure, the second nanostructure, and the third nanostructure.
(126) In accordance with some embodiments, a method for forming a semiconductor device structure is provided. The method includes providing a substrate having a base and a fin over the

base. The method includes forming a nanostructure stack over the fin. The nanostructure stack includes a first nanostructure, a second nanostructure, a third nanostructure, and a fourth nanostructure sequentially formed over the fin, and the second nanostructure is thicker than the fourth nanostructure. The method includes forming a gate stack over the nanostructure stack and the fin. The method includes partially removing the nanostructure stack and the fin, which are not covered by the gate stack, to form a trench in the nanostructure stack and the fin. The method includes removing end portions of the first nanostructure and the third nanostructure through the trench to form a first recess and a second recess in the nanostructure stack. The first recess is between the fin and the second nanostructure, and the second recess is between the second nanostructure and the fourth nanostructure. The method includes forming a first inner spacer and a second inner spacer in the first recess and the second recess respectively. The method includes forming a stressor structure in the trench and connected to the second nanostructure and the fourth nanostructure.

(127) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor device structure, comprising: providing a substrate having a base and a fin over the base; forming a nanostructure stack over the fin, wherein the nanostructure stack comprises a first nanostructure, a second nanostructure, a third nanostructure, and a fourth nanostructure sequentially formed over the fin, and the second nanostructure is thicker than the fourth nanostructure; forming a gate stack over the nanostructure stack and the fin; partially removing the nanostructure stack and the fin, which are not covered by the gate stack, to form a trench in the nanostructure stack and the fin; removing end portions of the first nanostructure and the third nanostructure through the trench to form a first recess and a second recess in the nanostructure stack, wherein the first recess is between the fin and the second nanostructure, and the second recess is between the second nanostructure and the fourth nanostructure; forming a first inner spacer and a second inner spacer in the first recess and the second recess respectively; and forming a stressor structure in the trench and connected to the second nanostructure and the fourth nanostructure, wherein the forming of the stressor structure comprises: forming a lightly doped layer in the trench to cover sidewalls of the first inner spacer, the second nanostructure, the second inner spacer, and the fourth nanostructure; and forming a heavily doped structure over the lightly doped layer, wherein the lightly doped layer and the heavily doped structure are doped with a dopant, a first concentration of the dopant in the lightly doped layer is less than a second concentration of the dopant in the heavily doped structure, and the heavily doped structure laterally overlaps with the second nanostructure.
2. The method for forming the semiconductor device structure as claimed in claim 1, wherein the second nanostructure is thicker than the first nanostructure.
3. The method for forming the semiconductor device structure as claimed in claim 1, wherein the second nanostructure is thicker than the third nanostructure.
4. The method for forming the semiconductor device structure as claimed in claim 1, further comprising: before forming the stressor structure in the trench, forming an insulating layer in the trench, and the stressor structure is formed over the insulating layer.

5. The method for forming the semiconductor device structure as claimed in claim 4, wherein the insulating layer and the first inner spacer are made of a same material.
6. The method for forming the semiconductor device structure as claimed in claim 4, wherein the stressor structure partially extends into the insulating layer.
7. The method for forming the semiconductor device structure as claimed in claim 1, wherein the stressor structure partially extends into the first inner spacer and the second inner spacer.
8. The method for forming the semiconductor device structure as claimed in claim 1, wherein a width of the second nanostructure increases toward the fin.
9. The method for forming the semiconductor device structure as claimed in claim 1, wherein a first distance between the second nanostructure and the fin is substantially equal to a second distance between the second nanostructure and the fourth nanostructure.
10. A method for forming a semiconductor device structure, comprising: providing a substrate having a base and a fin over the base; forming a nanostructure stack over the fin, wherein the nanostructure stack comprises a first nanostructure, a second nanostructure, a third nanostructure, and a fourth nanostructure sequentially formed over the fin; forming a gate stack over the nanostructure stack and the fin; partially removing the nanostructure stack and the fin, which are not covered by the gate stack, to form a trench in the nanostructure stack and the fin; removing end portions of the first nanostructure and the third nanostructure through the trench to form a first recess and a second recess in the nanostructure stack, wherein the first recess is between the fin and the second nanostructure, and the second recess is between the second nanostructure and the fourth nanostructure; forming a first inner spacer, a second inner spacer, and an insulating layer in the first recess, the second recess, and a bottom portion of the trench respectively, wherein the second nanostructure is thicker than the first inner spacer, a first top surface of the insulating layer is lower than a second top surface of the fin, the first top surface has a third recess, and the insulating layer is spaced apart from the first inner spacer and the second inner spacer; and forming a stressor structure in the trench and connected to the second nanostructure and the fourth nanostructure, wherein the stressor structure comprises a lightly doped layer and a heavily doped structure in the lightly doped layer, the lightly doped layer continuously covers sidewalls of the first inner spacer, the second nanostructure, the second inner spacer, and the fourth nanostructure, the lightly doped layer has a first portion and a second portion, the first portion covers the sidewall of the second nanostructure, the second portion covers the sidewall of the fourth nanostructure, the first portion is thicker than the second portion, the lightly doped layer is partially in the third recess of the first top surface of the insulating layer, and the heavily doped structure laterally overlaps with the second nanostructure.
11. The method for forming the semiconductor device structure as claimed in claim 10, wherein the second nanostructure is thicker than the second inner spacer.
12. The method for forming the semiconductor device structure as claimed in claim 10, wherein the fin is wider than the second nanostructure.
13. The method for forming the semiconductor device structure as claimed in claim 12, wherein the second nanostructure is wider than the fourth nanostructure.
14. The method for forming the semiconductor device structure as claimed in claim 10, wherein the first inner spacer is wider than the second inner spacer.
15. The method for forming the semiconductor device structure as claimed in claim 10, wherein the lightly doped layer and the heavily doped structure are doped with a dopant, a first concentration of the dopant in the lightly doped layer is less than a second concentration of the dopant in the heavily doped structure, the lightly doped layer has a first portion and a second portion, the first portion and the second portion have the same first concentration of the dopant, the first portion covers the sidewall of the second nanostructure, the second portion covers the sidewall of the fourth nanostructure, a first thickness of the first portion is greater than a second thickness of the second portion, and the first thickness and the second thickness are measured along a longitudinal axis of

the fin.

16. A method for forming a semiconductor device structure, comprising: providing a substrate; forming a nanostructure stack over the substrate, wherein the nanostructure stack comprises a first nanostructure, a second nanostructure, a third nanostructure, and a fourth nanostructure sequentially formed over the substrate, the second nanostructure is thicker than the first nanostructure, the second nanostructure is thicker than the third nanostructure, the second nanostructure is thicker than the fourth nanostructure, and a ratio of a first thickness of the second nanostructure to a second thickness of the fourth nanostructure ranges from about 1.5 to about 2; forming a gate stack over the nanostructure stack and the substrate; partially removing the nanostructure stack and the substrate, which are not covered by the gate stack, to form a trench in the nanostructure stack and the substrate; removing end portions of the first nanostructure and the third nanostructure through the trench to form a first recess and a second recess in the nanostructure stack, wherein the first recess is between the substrate and the second nanostructure, and the second recess is between the second nanostructure and the fourth nanostructure; forming a first inner spacer and a second inner spacer in the first recess and the second recess respectively; and forming a stressor structure in the trench and connected to the second nanostructure and the fourth nanostructure, wherein the stressor structure has a lightly doped portion and a heavily doped portion in the lightly doped portion, the lightly doped portion has a first part and a second part, the first part covers a first sidewall of the second nanostructure, the second part covers a second sidewall of the second inner spacer, the first part has a convex sidewall facing away from the second nanostructure, the second part has a concave sidewall facing away from the second inner spacer, and the heavily doped portion laterally overlaps with the second nanostructure.

17. The method for forming the semiconductor device structure as claimed in claim 16, wherein the second nanostructure has an upper portion and a lower portion, the upper portion is between the lower portion and the third nanostructure, and the upper portion is narrower than the lower portion.

18. The method for forming the semiconductor device structure as claimed in claim 16, wherein the lightly doped portion and the heavily doped portion are doped with a dopant, and a first concentration of the dopant in the lightly doped portion is less than a second concentration of the dopant in the heavily doped portion.

19. The method for forming the semiconductor device structure as claimed in claim 18, wherein the lightly doped portion extends into the first inner spacer.

20. The method for forming the semiconductor device structure as claimed in claim 16, further comprising: forming an insulating layer in a bottom portion of the trench in the nanostructure stack and the substrate, wherein the insulating layer has a concave curved top surface, the lightly doped portion is over the concave curved top surface of the insulating layer, the lightly doped portion has a first wavy sidewall and a second wavy sidewall opposite to the first wavy sidewall, and the concave curved top surface is connected between the first wavy sidewall and the second wavy sidewall.